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Multiple transition states in unimolecular reactions: A transition state switching model. Application to the $C_4H_8^+$ system

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A transition state switching model is developed for use in systems where more than one transition state occurs along the reaction coordinate. The model is cast in the perspective of both the unified statistical theory (UST) of Miller and of variational transition state theory. The basic assumptions are those common to transition state theory and RRKM-QET. A reaction branching analysis leads to reaction probabilities for a number of potential surfaces and appropriate expressions are delineated for both unimolecular and bimolecular reactions. The theory is developed from a microcanonical viewpoint and rigorously conserves both energy E and angular momentum J . Comparison is made with experimental data for the $C_4H_8^+$ system where absolute unimolecular rate constants and branching ratios have been measured as a function of energy, bimolecular rate constants, and branching ratios measured at room temperature (the ethylene ion-molecule reaction), the lifetime of $C_4H_8^+$ measured when formed by the ethylene ion-molecule reaction, and product kinetic energy distributions measured. The principal conclusions of the work are (i) a multiple transition state switching mechanism is required if the experimental data are to be adequately fit by theory; (ii) at least two transition states (i.e., dividing surfaces at points of minimum flux along the reaction coordinate) naturally occur in unimolecular reactions, one a tight (i.e., configurational) TS occurring near the unimolecular reactant and the second an orbiting (i.e., centrifugal) TS occurring near the bimolecular products; (iii) tight transition states naturally occur at energies $E^\ddagger$ less than the threshold energy for reaction E^0 when the system energy E is significantly greater than E^0 ; (iv) the relative abundance of the three product channels $C_2H_4^+ + C_2H_4$, $C_3H_5^+ + CH_3$, and $C_4H_7^+ + H$ are strong functions of both energy and angular momentum in the reactant $C_4H_8^+$ ion.

I. INTRODUCTION

In recent years a number of comparisons of RRKM-QET calculations with experimental data on the unimolecular decomposition of excited ions have led to conclusions that are in conflict with previously established interpretations of the dynamics of gas phase ionic processes. For example, RRKM-QET predictions are usually able to match measured unimolecular dissociation rates and branching ratios only when tight transition states are used in the calculations,^{1,2} and even then sometimes only when the activation energy is taken to be less than the zero-point energy difference between the unimolecular reactant and products.^{2(a)} From arguments based on microscopic reversibility these results are in conflict with the well established result that thermal bimolecular ion-molecule reactions often occur with rates approaching the capture rate, which, in turn, is determined by the much looser orbiting transition state^{1,2} on the long range potential. On the other hand, the measured product translational energy distributions of unimolecular reactions are often (but not always) matched by calculations based on the orbiting transition state.¹⁻³ Although this finding is in agreement with the thermal ion-molecule reaction data, it is in conflict with the absolute unimolecular rate and branching ratio results.

To the best of our knowledge no model has yet been

postulated which can resolve the apparent conflict summarized above. In this paper we suggest such a model. We show, using arguments based on Miller's unified statistical theory (UST),⁴ that the requirement that at least two transition states are operative along the reaction coordinate in a given reaction channel is not surprising and, in fact, may well be the general case. Specifically, the model we present here requires both a tight and orbiting transition state in each reaction channel and requires that the "best" transition state to use at each value of the system energy and angular momentum is the one of minimum flux.⁵

To test the "transition state switching model" presented here we apply it to the unimolecular decomposition of excited $C_4H_8^+$ ions. The $C_4H_8^+$ system is ideal for this purpose because of the large amount of experimental data which is currently available. Specifically, data on the absolute unimolecular dissociation rate have been obtained by Baer *et al.*⁶ using photoion-photoelectron coincidence (PIPECO) techniques and have recently been reinterpreted by Baer.⁷ Meisels *et al.*⁸ have used PIPECO to measure the product branching ratios. Data on the product translational energy distributions have been obtained by Holmes *et al.*⁹ and have recently been supplemented by more complete measurements by Bowers and co-workers.¹⁰ Data are also available on the thermochemistry of the system and on the reaction efficiency and product branching ratios for $C_4H_8^+$ formed from the thermal $C_2H_4^+ + C_2H_4$ ion-molecule reaction.¹¹ Furthermore, a number of isotope studies have also been reported.^{11,12}

This paper is organized as follows: We begin Sec. II

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with a brief review of the main assumptions of RRKM-QET, transition state theory (TST), variational TST, and UST. We then develop the transition state switching model from the perspective of UST. We also apply the model to double deep well potentials and discuss its role in the prediction of product translational energy distributions. In Sec. III we explain precisely how the model is applied to the $C_4H_8^+$ system and report the results of our calculations. A discussion of our results is given in Sec. IV. Finally, a number of appendices are included. The first summarizes Miller's derivation of the UST reaction probability and shows how this derivation must be altered to obtain the bimolecular reaction probability for a double deep well potential surface. The second discusses the bimolecular and unimolecular perspectives for chemical activation systems. The third summarizes the equations needed to calculate the fluxes that enter into the rate expressions. The fourth explains in detail the method we used to establish the parameters for the tight transition states.

II. THEORY

The transition state switching model can be developed either directly from the variational criterion for selecting the best transition state¹³ or from arguments based on Miller's more general UST.⁴ In this section we take the latter approach because it allows us to cast the model in a framework compatible with other multitransition state statistical-dynamical theories.¹⁴ We begin by briefly reviewing the fundamental concepts of TST, variational TST, and UST as they apply to unimolecular reactions. Then, in order, we develop the model, apply it to double well unimolecular potential surfaces, and discuss its implications in the calculation of product translational energy distributions.

A. Background

Consider the general unimolecular reaction



where $k(E, J)$ is the reaction rate constant at system energy E and angular momentum J . In classical mechanics the procedure for calculating $k(E, J)$ for a given potential surface involves two steps. The first is the generation of a set of initial conditions corresponding to the method of preparation of A.¹⁵ The second is the solution of the classical equations of motion for the trajectories corresponding to the selected initial conditions.¹⁶ The end result of this procedure is a lifetime distribution for A which can then be converted into $k(E, J)$ by standard methods.^{15,16}

The goal of statistical unimolecular rate theory (RRKM-QET) is to determine $k(E, J)$ in a way that avoids the difficult procedure of sampling initial conditions and the time consuming procedure of solving the equations of motion for a large ensemble of trajectories. This is accomplished by introducing two assumptions.^{16,17} The first is the quasiequilibrium or energy randomization hypothesis, which states that subject to known conservation laws all reactant states are populated uniformly and independently of the method of initial preparation.

The second is the fundamental assumption of TST. This assumption states that there exists somewhere in the system configuration space (or, more generally, in the system phase space) a dividing surface or transition state such that all reactive trajectories cross this surface once and only once, and all nonreactive trajectories never cross the surface.

The second assumption stated above allows the reactive flux to be taken as the appropriate local flux across the transition state surface. The first assumption states that the appropriate local flux is the equilibrium local flux. Taken together, these two assumptions lead directly to the following expression for $k(E, J)$:

$$k(E, J) = F^\ddagger(E, J) / \rho(E, J), \quad (\text{II.2})$$

where $F^\ddagger(E, J)$ is the local equilibrium flux through the transition state and $\rho(E, J)$ is the density of states of the reactant. Traditionally, Eq. (II.2) is usually written as

$$k(E, J) = N^\ddagger(E, J) / h\rho(E, J), \quad (\text{II.3})$$

where $N^\ddagger(E, J)$ is the sum of states available to the system at the transition state and h is Planck's constant.

It is often possible, in principle, to determine a transition state for which the fundamental assumption is true.¹⁷⁻¹⁹ The procedure for doing so is called the dynamical criterion for constructing a transition state.¹⁸ However, for systems involving many degrees of freedom, it is difficult to apply the dynamical criterion. Furthermore, it is known that under certain conditions a single transition state for which the fundamental assumption is true may not even exist.^{14,18,19} For these situations variational TST can be employed. The variational criterion for constructing a transition state is based on the fact that if some trajectories cross the transition state more than once, the resulting local flux must be an upper bound to the true reactive flux. Therefore, in variational TST the local flux is calculated as a function of the position of the transition state along the reaction coordinate. The best transition state is then chosen to be at the position for which the local flux is a minimum.

It should be noted here that according to either the dynamical or variational criterion the location of the transition state may vary with the system energy and angular momentum. Furthermore, if a transition state exists for which the fundamental assumption is true, both criteria successfully choose it to be the best transition state.^{18,19}

Miller's UST,⁴ which was developed to resolve some conflicts between rate constant expressions for direct vs complex bimolecular reactions, is a statistical-dynamical theory for situations in which there are two transition states or minima in the local flux along the reaction coordinate between reactant(s) and products. (Actually, it would be more appropriate to call the transition states dividing surfaces¹⁹ since trajectories can cross them more than once.) The goal of UST is to accurately describe the true system dynamics within the "interaction region" between the transition states using only the system properties at the transition states and at the maximum in the local flux (which must exist at

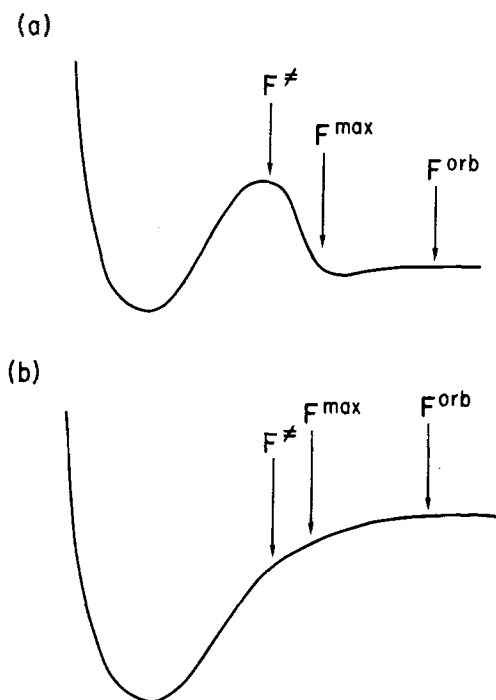


FIG. 1. Schematic diagrams of reaction coordinates containing a tight transition state. (a) The tight transition is located at a saddle point, or maximum, on the potential energy surface; (b) there is no saddle point on the potential energy surface—the tight transition state corresponds to the point of minimum flux.

some point between the two minima). For unimolecular reactions the resulting expression for $k(E, J)$ is

$$k(E, J) = F_1^\ddagger(E, J) \omega(E, J) / \rho(E, J), \quad (\text{II. 4})$$

with

$$\omega(E, J) = \frac{F_2^\ddagger(E, J)}{F_1^\ddagger(E, J) + F_2^\ddagger(E, J) - \frac{F_1^\ddagger(E, J)F_2^\ddagger(E, J)}{F^{\max}(E, J)}}, \quad (\text{II. 5})$$

where $F_1^\ddagger(E, J)$ and $F_2^\ddagger(E, J)$ are the local fluxes at the first and second transition states (as the system goes from reactants to products), respectively, and $F^{\max}(E, J)$ is the maximum in the local flux in the interaction region. [For bimolecular reactions Eq. (II. 4) remains unchanged, except $\rho(E, J)$ now becomes the density of states of the reactant pair per unit volume.]

B. The transition state switching model

Within the context of UST the aim of the transition state switching model is to approximate $\omega(E, J)$ in such a way that the largely unknown quantity $F^{\max}(E, J)$ is not needed. We begin by considering the general unimolecular dissociation given by Eq. (II. 1) and assume that the corresponding reaction coordinate is characterized by two transition states, an orbiting transition state¹⁷ located at the centrifugal barrier near the products region of the potential energy surface and a tight transition state located at some point closer to the unimolecular reactant. If there is a maximum in the potential energy surface along the reaction coordinate, as in Fig. 1(a), then this maximum, or saddle point, is usually taken as the location of the tight transition state. However, even for systems which are not characterized by

saddle points along the reaction coordinate [see Fig. 1(b)], it is logical to expect a minimum in the local flux and hence a second transition state somewhere between the orbiting transition state and the unimolecular reactant. This transition state is expected for the following reason: On moving along the reaction coordinate from reactant to products, changes in the flux arise primarily from two sources: the conversion of nonfixed energy (available for randomization) into potential energy; and the conversion of vibrational degrees of freedom of the reactant into rotational and translational degrees of freedom of the products. The former process tends to decrease the flux, while the latter tends to increase it. The net result of this competition is to produce a minimum in the flux (or, in canonical terms, a maximum in the local free energy). If $F^\ddagger(E, J)$ represents the flux at the tight transition state and $F^{\text{orb}}(E, J)$ represents the flux through the orbiting transition state, then Eq. (II. 5) can be rewritten as

$$\omega = \frac{F^{\text{orb}}}{F^\ddagger + F^{\text{orb}} - \frac{F^\ddagger F^{\text{orb}}}{F^{\max}}}, \quad (\text{II. 6})$$

where the arguments have been dropped for notational clarity.

It is useful to approximate Eq. (II. 6) in a way that eliminates $F^{\max}$. This can be done for a potential surface which increases continuously from reactant to products, such as in Fig. 1(b), in the following limiting situations: (1) energies close to the reaction threshold and (2) energies significantly greater than the reaction threshold. At energies close to the reaction threshold F^{orb} will be much smaller than either $F^\ddagger$ or $F^{\max}$, because the nonfixed energy at the orbiting transition state will be much smaller than that at the tight transition state or at the location corresponding to $F^{\max}$. As a first approximation $F^{\max}$ can be taken to be equal to $F^\ddagger$, yielding

$$\omega \xrightarrow{E \rightarrow \text{threshold}} \frac{F^{\text{orb}}}{F^\ddagger}, \quad (\text{II. 7})$$

as illustrated in Fig. 2(a). At energies significantly greater than the reaction threshold the decrease in the nonfixed energy at the orbiting transition state will become negligible when compared to the total nonfixed energy. Therefore, since there are a larger number of translational and rotational degrees of freedom available to the orbiting transition state, F^{orb} will be larger than $F^\ddagger$. Hence as a first approximation $F^{\max}$ can be set equal to F^{orb} , giving

$$\omega \xrightarrow{E \gg \text{threshold}} 1, \quad (\text{II. 8})$$

as is illustrated in Fig. 2(b).

Although the specific arguments given above have considered only energy, similar reasoning can be used for the variations of the relative values of the fluxes with respect to J . The important point to note is that F^{orb} usually rises much more rapidly with energy than $F^\ddagger$.^{1,2} Therefore, there will usually be a relatively small (E, J) range for which the magnitudes of F^{orb} and $F^\ddagger$ are comparable. If this range is small enough, then the error introduced by assuming

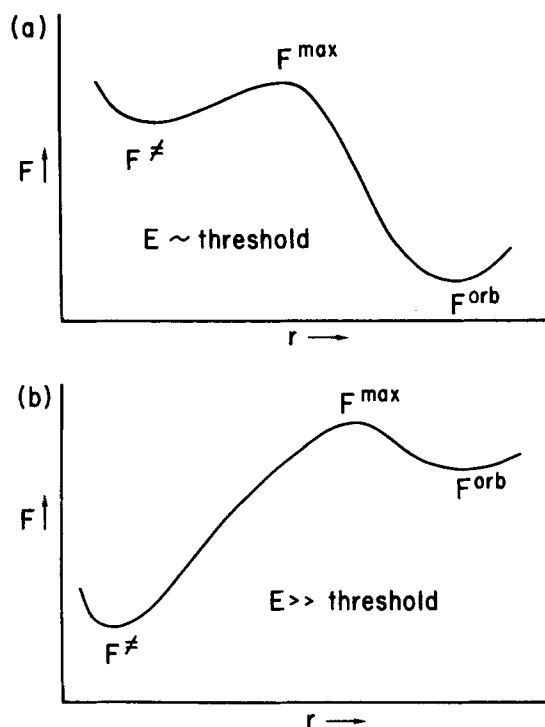


FIG. 2. Schematic diagrams showing the relative values of the fluxes for a potential energy surface such as that in Fig. 1(b). (a) For energies close to threshold; (b) for energies much greater than threshold.

$$F^{\max} = \text{greater of } \{F^{\ddagger}, F^{\text{orb}}\} \quad (\text{II. 9})$$

for all (E, J) will be small. Hence, in order to avoid the difficult problem of calculating $F^{\max}$, we assume that Eq. (II. 9) is always true. By combining Eqs. (II. 4), (II. 6), and (II. 9), the final result for $k(E, J)$ can be written as

$$k(E, J) = F^{\min}(E, J) / \rho(E, J), \quad (\text{II. 10})$$

where

$$F^{\min} = \text{lesser of } \{F^{\ddagger}, F^{\text{orb}}\}. \quad (\text{II. 11})$$

Note that these results also follow directly from the variational criterion that the best transition state is the one of minimum flux.¹³ Furthermore, they also apply to potential surfaces for which there is a saddle point along the reaction coordinate and a very shallow well between the saddle point and products, such as in Fig. 1(a) [note that for the surface in Fig. 1(a), $F^{\min}$ always equals $F^{\ddagger}$, because zero-point energy at the saddle point is larger than the zero-point energy of the products]. If, on the other hand, the well between the saddle point and products is deep, then the procedures discussed in Sec. II C are more appropriate.

Equation (II. 11) can result in a $k(E, J)$ which is not a smooth function of E and J (this may be an unavoidable consequence of variational TST under certain conditions). Smooth behavior can be returned to $k(E, J)$ by using UST directly. However, for systems of physical interest, this involves the difficult problem of constructing a potential energy surface and locating the various extrema in the local flux. Furthermore, application of the model to experimental data will generally involve

averaging $k(E, J)$ over distributions in E and J . This averaging will tend to prevent nonsmooth behavior from appearing in the final results.

C. Application to double well potentials

A schematic reaction coordinate diagram for a double well unimolecular potential energy surface is given in Fig. 3, along with the definitions of the relevant fluxes and densities of states. Using the switching model just developed, the flux from the second, more shallow, well toward products is given by

$$F^{\min} = \text{lesser of } \{F_2^{\ddagger}, F^{\text{orb}}\}. \quad (\text{II. 12})$$

The overall rate constant for reaction from the first, deeper well to products is, according to UST,

$$k = (F_1^{\ddagger} / \rho_1) \omega, \quad (\text{II. 13})$$

where ω is given by

$$\omega = \frac{F^{\min}}{F_1^{\ddagger} + F^{\min} - \frac{F_1^{\ddagger} F^{\min}}{F_1^{\max}}}, \quad (\text{II. 14})$$

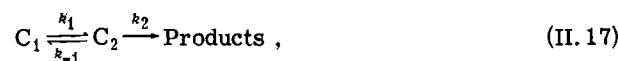
which is analogous to Eq. (II. 6). Since $F_1^{\max}$ corresponds to a flux through a surface in a potential well, it is reasonable as a first approximation to assume $F_1^{\max}$ is much greater than $F^{\min}$ and $F_1^{\ddagger}$ due to the large amount of non-fixed energy available at the well (the situation in which the well is shallow has been considered by Miller⁴ and is also discussed in Sec. II B above). In this limit Eq. (II. 14) reduces to the standard expression for reaction through an intermediate complex⁴

$$\omega = F^{\min} / (F_1^{\ddagger} + F^{\min}). \quad (\text{II. 15})$$

When Eq. (II. 15) is combined with Eq. (II. 13), the expression for the rate constant becomes

$$k = F_1^{\ddagger} F^{\min} / [\rho_1 (F_1^{\ddagger} + F^{\min})]. \quad (\text{II. 16})$$

It is important to note that, for the kinetic scheme corresponding to the surface in Fig. 3,



the rate constant given by Eq. (II. 13), using Eq. (II. 15)

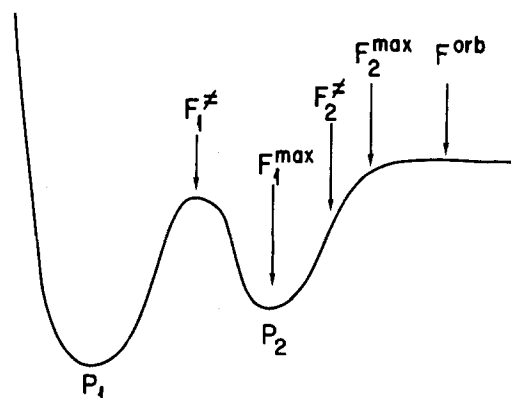


FIG. 3. Schematic diagram of a unimolecular reaction coordinate containing two wells; ρ_1 and ρ_2 are densities of states at the indicated wells, $F_1^{\ddagger}$ and $F_2^{\ddagger}$ are fluxes through tight transition states, and F^{orb} is the flux at the orbiting transition state.

for ω , applies to the rate of conversion of C_1 to products. This quantity is often difficult to measure experimentally since C_1 and C_2 are not always distinguishable. For example, in PIPECO experiments the rate of product formation is measured in terms of the total concentration of C_1 and C_2 , and the observed rate expression is

$$[d(\text{Products})]/dt = k_{\text{obs}}([C_1] + [C_2]). \quad (\text{II. 18})$$

A steady state analysis for C_2 yields

$$k_{\text{obs}} = k_1 k_2 / (k_1 + k_2 + k_{-1}). \quad (\text{II. 19})$$

Inserting

$$k_1 = F_1^\ddagger / \rho_1, \quad (\text{II. 20a})$$

$$k_{-1} = F_1^\ddagger / \rho_2, \quad (\text{II. 20b})$$

$$k_2 = F^{\text{min}} / \rho_2 \quad (\text{II. 20c})$$

yields

$$k_{\text{obs}} = F_1^\ddagger F^{\text{min}} / [F_1^\ddagger (\rho_1 + \rho_2) + \rho_1 F^{\text{min}}]. \quad (\text{II. 21})$$

Thus, for experiments of this type both ρ_1 and ρ_2 enter into the observed rate constant. In the limit that $\rho_2 \ll \rho_1$, which implies that the outer well is much shallower than the inner well [but is still deep enough for Eq. (II. 15) to apply], Eq. (II. 21) reduces to Eq. (II. 16) and k_{obs} may be evaluated without calculating ρ_2 .

This completes our discussion of the application of the switching model to double well unimolecular reactions. In Appendix A double deep well bimolecular reactions are discussed in more detail. In particular, we derive an expression for the bimolecular reaction probability using an approach based on a probability branching analysis. This method illustrates clearly the relation of the underlying assumptions stated earlier in this section to the assumptions of UST. The resulting expression for the reaction probability is given by

$$\omega = \frac{F_2}{F_1 + F_2 + \frac{F_1 F_2}{F_x}}, \quad (\text{II. 22})$$

where F_1 is the minimum flux between reactants and the first well, F_2 is the minimum flux between the second well and the products, and F_x is the flux at the saddle point between the two wells.

D. Product translational energy release distribution

For systems described by a reaction coordinate with multiple transition states the product translational energy distribution depends upon the relative positions of the two transition states. If the tight transition state occurs late along the reaction coordinate, then the Marcus (statistical adiabatic) theory²⁰⁻²² of translationally driven reactions may be applicable. This theory predicts relatively large values of product translational energy in accordance with the principle of microscopic reversibility.

If the tight transition state occurs early along the reaction coordinate, then it may well be that randomization of energy among all the degrees of freedom, including those which eventually contribute to the product translational energy, will continue to occur after the

system passes the tight transition state. If this occurs, the product translational energy distribution will be governed by the statistical distribution of states at the orbiting transition state, regardless of which of the two transition states is rate determining. Thus, in this case, the probability of a given translational energy release is given by¹⁻³

$$P(E, J; E_t) dE_t = F^{\text{orb}}(E, J; E_t) dE_t / F^{\text{orb}}(E, J), \quad (\text{II. 23})$$

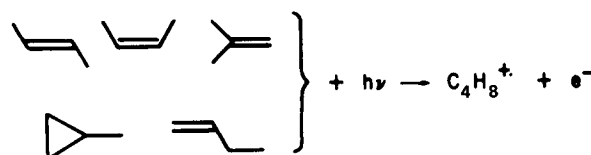
where $F^{\text{orb}}(E, J; E_t)$ is the orbiting flux per unit translational energy at translational energy E_t .

III. APPLICATION TO THE $C_4H_8^+$ SYSTEM

The switching model developed in the previous section provides the framework for treating reaction channels containing at least two transition states (one tight and one orbiting) with or without a stable intermediate between them. In this section we apply the model to the decomposition of excited $C_4H_8^+$ ions. We begin by using some of the experimental data to construct a reasonable potential surface and set of transition states for the $C_4H_8^+$ system. Then, after summarizing the necessary rate expressions, we use the model to predict the energy dependence of the total absolute unimolecular decomposition rate constant and of the unimolecular product branching ratios, the thermal bimolecular branching ratios and reaction efficiency, the complex lifetime for $C_4H_8^+$ formed from $C_2H_4^+ + C_2H_4$, and the unimolecular product translational energy distributions.

A. The potential surface and transition states

In order to construct a reasonable potential surface and set of transition states for $C_4H_8^+$ we need to extract some preliminary information from some of the experiments. Baer *et al.*^{6,7} have used PIPECO techniques to measure the absolute rate constant for total loss of $C_4H_8^+$ generated by photoionization of each of the following five isomers:



(III. 1)

These workers found identical decomposition rates for the five isomers over the entire energy range sampled (0–0.5 eV above threshold for reaction), indicating that dissociation must be preceded by isomerization to a common structure (or set of structures). In their original analysis⁶ they reported a lifetime distribution given by a sum of two exponentials, suggesting that dissociation proceeds from two different isomeric structures. A subsequent reanalysis,⁷ however, has shown that their data are consistent with a single lifetime distribution and thus the formation of a single structure preceding all fragmentations.

Meisels *et al.*,⁸ using PIPECO techniques, reported the breakdown graphs for all six stable C_4H_8 isomers [those given in Eq. (III. 1), plus cyclobutane]

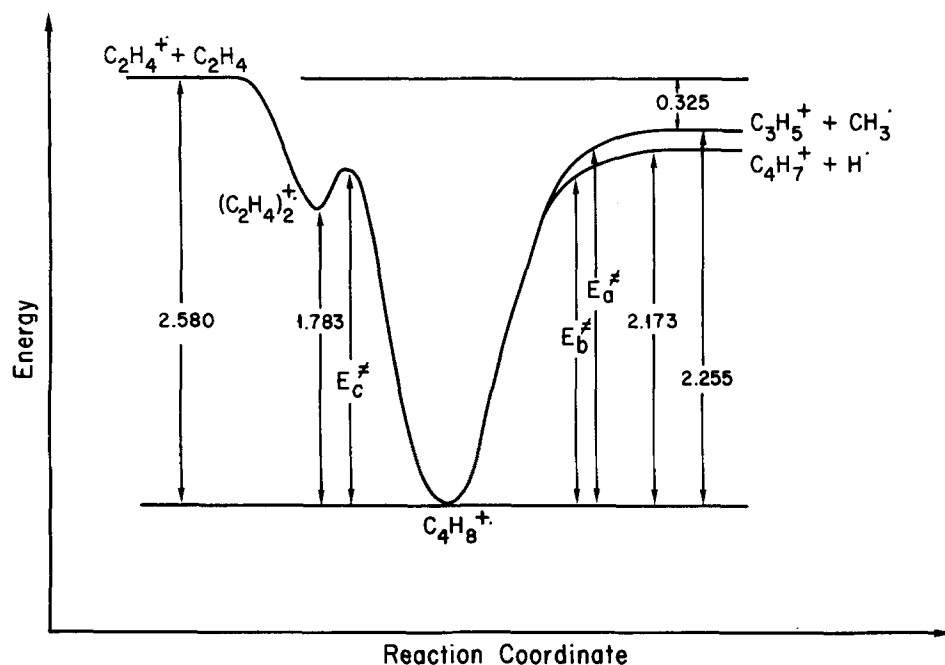
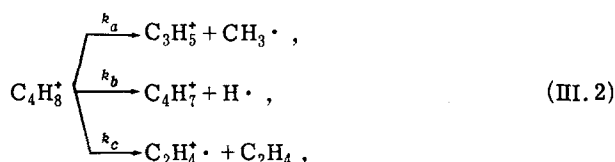


FIG. 4. Schematic potential energy surface used in the calculations presented here for the $C_4H_8^+$ system.

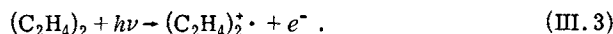


for energies 0–1 eV above threshold. The four non-cyclic isomers gave identical breakdown graphs indicating isomerization to a common isomer before fragmentation. Slightly different results were obtained for the two cyclic isomers $\triangleright$ and $\square$ but it is not clear whether the differences are experimentally significant. Hence, these two systems will not be discussed further here.

The product translational energy distributions for the $C_3H_5^+$ ^{9,10} and $C_4H_7^+$ ¹⁰ channels have recently been measured for all six neutral C_4H_8 isomers. Each of the reactions for the six isomers was observed at very high resolution (on the order of 15 000).¹⁰ Essentially the same distributions were obtained from all six isomers, although minor reproducible variations did occur in the $C_3H_5^+$ distributions. In no instance was any structure apparent in any of the metastable peaks. Furthermore, none of the resulting translational energy distributions showed evidence of substantial reverse activation energy.

The three sets of experiments just discussed suggest that rapid isomerization to a single isomeric structure occurs before dissociation. (The results also imply that energy randomization occurs before dissociation. This point will be discussed in detail in Sec. IV.) Hence, in the construction of a potential surface the $C_4H_8^+$ parent is represented by a single deep well corresponding to the most stable isomer. Also, since the $C_3H_5^+$ and $C_4H_7^+$ translational energy distributions show no evidence of reverse activation energy, the $J=0$ potential surface in each of these channels is taken to be a smooth, monotonically increasing function of the dis-

tance along the reaction coordinate toward products. For the $C_2H_4^+$ channel we note that Ceyer *et al.*²³ have recently observed the loosely bound ionic dimer $(C_2H_4)_2^+$ via the following photoionization reaction:



Therefore, we assume that the potential surface in the $C_2H_4^+$ channel has a well corresponding to the ionic dimer. The resulting potential surface is given in Fig. 4.

Two transition states are assumed to exist in the $C_3H_5^+$ and $C_4H_7^+$ channels, one tight and one orbiting. Justification for the existence of these transition states was discussed on a conceptual level in Sec. II. Here we also note that the large body of experimental data on thermal ion–molecule reaction rates shows conclusively that the upper bound to thermal, heavy particle transfer rates is given, within experimental error, by the collision rate determined from the long range potential.^{24–26} This collision rate is, in turn, governed by the orbiting transition state. On the other hand, previous comparisons of RRKM–QET calculations with PIPECO absolute unimolecular rate data have shown that when only orbiting transition states are used the calculated rates are orders of magnitude greater than the experimental data.^{1–2} However, tight transition states give a substantial improvement between experiment and theory.^{1,2(a)}

In the $C_2H_4^+$ channel both an orbiting and tight transition state are assumed to exist. However, in this case we place the tight transition state at the saddle point between the $C_2H_4^+$ deep well complex and $(C_2H_4)_2^+$ dimer and the orbiting transition state between the dimer and products.

The choice of values for the various transition state parameters is discussed in detail in Appendix D. We would like to point out here that, although a number of these parameters are adjustable, there is sufficient

experimental data so that all parameters may be determined without using all of the experimental data. Therefore, the theoretical calculations pertaining to the remaining experimental results are done with essentially no adjustable parameters.

B. The rate expressions

Before discussing the theoretical calculations corresponding to specific experiments it is useful to first summarize the rate expressions used in the calculations. For notational clarity in the remainder of this paper, the subscripts a , b , and c are used to represent the $C_3H_5^+$, $C_4H_7^+$, and $C_2H_4^+$ channels, respectively. The difference in zero-point energy between the products in channel ($j = a, b, c$) and the deep well $C_4H_8^+$ structure is denoted by E_j^0 ; the corresponding difference between the tight transition state in channel j and the deep well $C_4H_8^+$ structure is denoted by $E_j^\ddagger$. These energies are illustrated in Fig. 5.

Consider the three possible unimolecular processes originating at the deep $C_4H_8^+$ well, as given in Eq. (III.2). For given values of the total system energy and angular momentum each of the rate constants in the reaction scheme can be expressed in the form

$$k_j(E, J) = F_j'(E, J) / \rho(E, J), \quad (\text{III. 4})$$

where $F_j'(E, J)$ is the reactive flux from $C_4H_8^+$ to products in channel j , and $\rho(E, J)$ is the density of states at the $C_4H_8^+$ well. The three reactive fluxes, within the context of the switching model presented in Sec. II, are given by [see Eqs. (II.11), (II.13), and (II.14)]

$$F_a' = \text{lesser of } \{F_a^{\text{orb}}, F_a^\ddagger\}, \quad (\text{III. 5a})$$

$$F_b' = \text{lesser of } \{F_b^{\text{orb}}, F_b^\ddagger\}, \quad (\text{III. 5b})$$

$$F_c' = F_c^{\text{orb}} F_c^\ddagger / (F_c^{\text{orb}} + F_c^\ddagger), \quad (\text{III. 5c})$$

where the arguments E and J have been omitted for notational clarity. Equations (III.5), together with Eq. (II.23) for translational energy release, provide the essential framework for treating unimolecular processes, i.e., those which originate at the $C_4H_8^+$ deep well.

For processes involving the thermal ethylene ion-ethylene molecule reaction the system originates as separated reactants in channel c . The flux from the reactant region to the first well (the van der Waals complex) is, according to the assumptions discussed above, given by F_c^{orb} , the flux at the orbiting transition state. The probability that such a collision will result in reactive dissociation into channel a or b is given by the expression [see Eq. (II.22) and Appendix A]

$$P_j = \frac{F_j'}{F_a' + F_b' + F_c^{\text{orb}} + \frac{(F_a' + F_b') F_c^{\text{orb}}}{F_c^\ddagger}}, \quad (\text{III. 6})$$

where $j = a$ or b . Therefore, the thermal bimolecular rate constant for reaction into channel a or b is

$$k_j(T) = F_j(T) / Q_c, \quad (\text{III. 7a})$$

where Q_c is the partition function of $C_2H_4^+ + C_2H_4$ per unit volume, and where the thermal reactive flux is given by

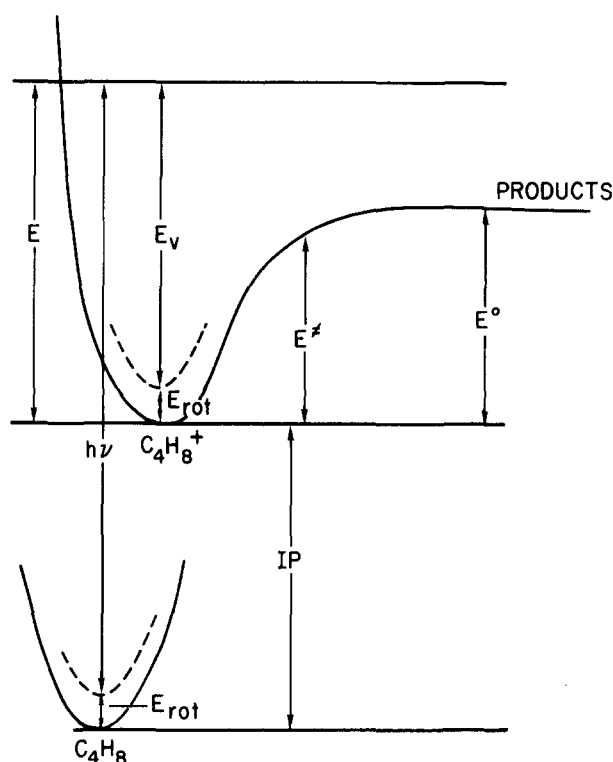


FIG. 5. Schematic diagram showing definitions of energies used describing photoionization of C_4H_8 . Note that C_4H_8 is assumed to be in the ground vibrational level, and E_{rot} is assumed not to change upon ionization.

$$F_j(T) = \int_0^\infty dE_R \exp(-E_R/k_B T) \times \int_0^{J_{\text{max}}(E_R)} dJ 2J F_c^{\text{orb}}(E_R, J) P_j(E_R, J) / h. \quad (\text{III. 7b})$$

In Eq. (III.7b) $J_{\text{max}}(E_R)$ is the maximum J for which a $C_2H_4^+ + C_2H_4$ orbiting collision can occur (see Appendix C) and

$$E_R = E - E_c^0 \quad (\text{III. 7c})$$

is the total nonfixed energy of the $C_2H_4^+ + C_2H_4$ reactants.

One more quantity which is of interest is the bimolecular flux from the $C_2H_4^+ + C_2H_4$ region to the deep $C_4H_8^+$ well. By the principle of microscopic reversibility, this flux is equivalent to the flux from the well to the $C_2H_4^+ + C_2H_4$ fragments, which is F_c' as given in Eq. (III.5c).

The actual equations used to evaluate the necessary fluxes and densities are given in Appendix C.

C. The absolute unimolecular rate constant for dissociation of $C_4H_8^+$

In the PIPECO experiments of Baer *et al.*,^{6,7} a thermal distribution of C_4H_8 molecules is ionized by a photon beam via the reaction



Assuming (1) that the vibrational energy of the neutral is negligible, (2) that there is no transfer of angular

momentum in the ionization process, and (3) that the moment of inertia of C_4H_8 does not change significantly upon ionization, the vibrational energy of the resulting ion is given by²⁷

$$E_v = h\nu - I.P., \quad (III. 9a)$$

where $h\nu$ is the photon energy and I.P. is the ionization potential of C_4H_8 (see Fig. 5). Hence, according to these assumptions the total system energy is (see Fig. 5)

$$E = E_v + BJ^2, \quad (III. 9b)$$

where B is the rotational constant of $C_4H_8^{+28}$. Although these assumptions don't hold exactly, they are sufficient approximations for the type of comparisons intended in this work.

PIPECO absolute rate experiments measure the rate of formation of a given product ion as a function of the fragmentation time t .^{29(a)} In the $C_4H_8^+$ experiments of Baer *et al.*⁶ the rate of formation of the $C_3H_5^+$ product ion was measured. In Baer's reanalysis⁷ of the original data he assumed that reaction occurs via only one pathway. He therefore fit the measured data on $d[C_3H_5^+]/dt$ to the expression

$$\frac{d[C_3H_5^+]}{dt} = [C_4H_8^+]_0 k_a^{obs}(E_v) \exp[-k_{tot}^{obs}(E_v)t], \quad (III. 10)$$

where

$$k_{tot}^{obs}(E_v) = \sum_{j=a,b,c} k_j^{obs}(E_v), \quad (III. 11)$$

where $k_{tot}^{obs}(E_v)$ and $k_j^{obs}(E_v)$ are the reported rate constants. Actually, since only relative values of $d[C_3H_5^+]/dt$ are measured, only $k_{tot}^{obs}(E_v)$ can be determined without additional knowledge on the energy dependence of the branching ratio.

In order to determine the relation between $k_{tot}^{obs}(E_v)$ and the microscopic rate constants, Eq. (III. 10) has to be rewritten in terms of these latter quantities. This leads to

$$\frac{d[C_3H_5^+]}{dt} = [C_4H_8^+]_0 \times \int_0^\infty dJ P(J) k_a(E_v, J) \exp[-k_{tot}(E_v, J)t], \quad (III. 12)$$

where $P(J)$ is the initial thermal distribution of J states given (classically) by

$$P(J) = (2J)^2 \exp(-BJ^2/k_B T) / \int_0^\infty dJ (2J)^2 \exp(-BJ^2/k_B T), \quad (III. 13)$$

and where

$$k_{tot}(E_v, J) = \sum_{j=a,b,c} k_j(E_v, J). \quad (III. 14)$$

Equating the time derivatives of the right-hand sides of Eqs. (III. 10) and (III. 12) and assuming that $\partial k_{tot}^{obs}(E_v)/\partial t = 0$ [implicitly assumed in Eq. (III. 10)], and that $\partial k_a^{obs}(E_v)/\partial t = 0$, and solving for $k_{tot}^{obs}(E_v)$ gives

$$k_{tot}^{obs}(E_v) = \frac{\int_0^\infty dJ P(J) k_a(E_v, J) k_{tot}(E_v, J) \exp[-k_{tot}(E_v, J)t]}{\int_0^\infty dJ P(J) k_a(E_v, J) \exp[-k_{tot}(E_v, J)t]}. \quad (III. 15)$$

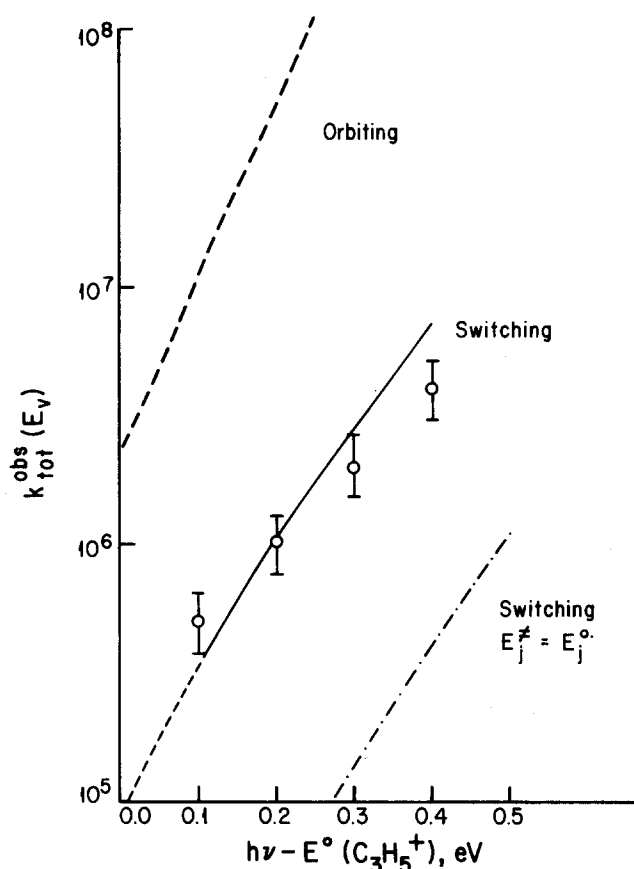


FIG. 6. Apparent absolute unimolecular dissociation rate constant for $C_4H_8^+$ as determined in PIPECO experiments, as a function of vibrational energy above the $C_3H_5^+ + CH_3 \cdot$ threshold. Experimental points from Ref. 7; —, theoretical results using the transition state switching model presented here; ----, theoretical results using only orbiting transition states in each channel; - · - · -, theoretical results using the transition state switching model with $E_j^* = E_j^0$ in each channel.

Note that $k_{tot}^{obs}(E_v)$ predicted by Eq. (III. 15) is time dependent^{29(b),(c)} (the exponential term causes J states with faster dissociation rates to have smaller relative populations as time increases). There is no unambiguously defined value of t which should be used in Eq. (III. 15) for comparison with experiment. Thus, we have used Eq. (III. 15) to calculate $k_{tot}^{obs}(E_v)$ vs E_v for times corresponding to the lower and upper limits of the experimental time regime. The results at these two limits were found to differ by less than 7% for all energies of interest here. This supports the experimental evidence that the time dependence of the reported rate constants is small in the cases discussed here.

A comparison between the theoretical total rate constant calculated from Eq. (III. 15) using an average time and the experimental results is shown in Fig. 6. The theoretical calculations were performed using $E_j^* < E_j^0$ for each channel. The reasons for doing so were discussed on a conceptual level in Sec. II and are discussed in more detail in Sec. IV. Also plotted in this figure are the theoretical results obtained by using only the orbiting transition state in each channel and by using the switching model with $E_j^* = E_j^0$ for each channel. Note that the three theoretical curves in this figure do not con-

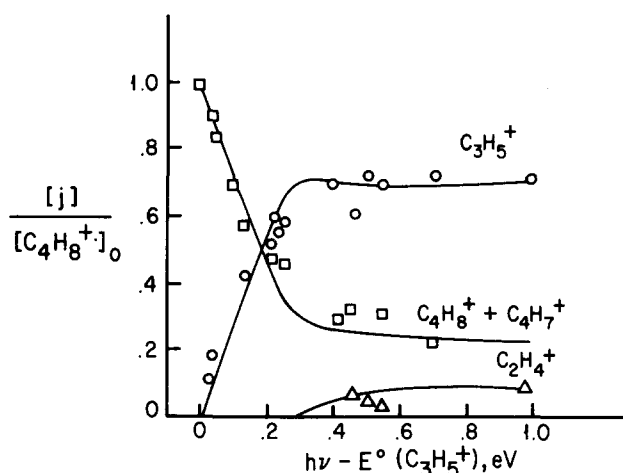


FIG. 7. Relative unimolecular product ion intensities as a function of vibrational energy above the $C_3H_5^+ + CH_3 \cdot$ threshold. Experimental points from Ref. 8; solid lines represent the calculations presented here.

verge to the same value at threshold. This occurs because of the averaging defined in Eq. (III.15).

D. The unimolecular branching ratio

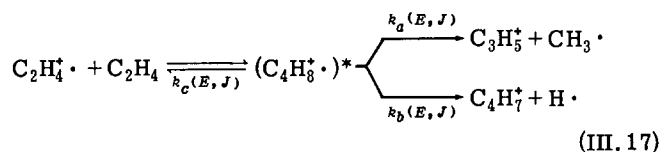
The PIPECO data obtained by Meisels *et al.*^{8,30} on the branching ratio for Reaction (III.2) are plotted in Fig. 7. The theoretical branching ratios can be obtained by integrating Eq. (III.12) from $t=0$ to $t=\tau$. This leads to

$$\frac{[j]}{[C_4H_8^+]_0} = \int_0^\infty dJ P(J) \frac{k_j(E_v, J)}{k_{tot}(E_v, J)} \times \{1 - \exp[-k_{tot}(E_v, J)\tau]\}. \quad (III.16)$$

This equation was used to calculate the solid lines in Fig. 7. In the calculations the assumptions leading to Eq. (III.9a) were used. The experimental time scale is estimated to be⁸ $\tau = 1.4 \times 10^{-6}$ sec with an uncertainty of $\pm 20\%$.

E. The bimolecular ethylene ion-molecule reaction

The reaction



has been studied by a variety of techniques and is known to proceed through a long lived complex at low reactant translational energies.^{11,12,31-33} Bowers *et al.*¹¹ have used ICR spectrometry to determine the thermal branching ratio between $C_4H_7^+$ and $C_3H_5^+$ and the overall thermal reaction efficiency for the conversion of $C_2H_4^+$ to products. Field³² has used mass spectrometry measurements to obtain an estimate of the lifetime of the intermediate complex.

Since an ICR experiment measures rate constants, the thermal branching ratio is given by the ratio of the thermal bimolecular rate constants for $C_4H_7^+$ and $C_3H_5^+$ formation³⁴

$$\frac{[C_4H_7^+]}{[C_3H_5^+]} = \frac{k_b(T)}{k_a(T)}. \quad (III.18)$$

Therefore, from Eq. (III.7a)

$$\frac{[C_4H_7^+]}{[C_3H_5^+]} = \frac{F_b(T)}{F_c(T)}. \quad (III.19)$$

The thermal reaction efficiency is defined by

$$P_{rxn}(T) = (k_a + k_b)/k_L, \quad (III.20a)$$

where k_L , the Langevin capture rate constant,²⁴⁻²⁶ can be written in terms of F_c^{orb} as

$$k_L = F_c^{orb}(T)/Q_c, \quad (III.20b)$$

where

$$F_c^{orb}(T) = \int_0^\infty dE_R \exp(-E_R/k_B T) \times \int_0^{J_{max}(E)} dJ 2J F_c^{orb}(E_R, J). \quad (III.20c)$$

Hence,

$$P_{rxn}(T) = [F_b(T) + F_c(T)]/[F_c^{orb}(T)]. \quad (III.21)$$

Note that Eq. (III.20a) offers no saving in computer time over Eq. (III.21), because even though k_L is a simple, well known expression, k_a and k_b require the calculation of Q_c .

In order to calculate the lifetime of the deep well $C_4H_8^+$ complex, all fragmentations proceeding from the $C_4H_8^+$ well must be viewed as unimolecular in nature, with the collision serving to activate the $C_4H_8^+$ species. From the reaction scheme in Eq. (III.17), the steady-state distribution function for chemically activated $C_4H_8^+$ is given by

$$P_{C_4H_8^+}(E_R, J) = \frac{2J \exp(-E_R/k_B T) \frac{F'_c(E_R, J)}{k_{tot}(E_R, J)}}{\int_0^\infty dE_R \exp(-E_R/k_B T) \int_0^{J_{max}} dJ 2J \frac{F'_c(E_R, J)}{k_{tot}(E_R, J)}} \quad (III.22)$$

where $k_{tot}(E_R, J)$ is given by Eq. (III.14) with E_v replaced by E_R . The thermally averaged rate constant for unimolecular dissociation into each of the three fragment channels is determined by averaging each $k_j(E_R, J)$ over the steady-state distribution function in Eq. (III.22).

This yields

$$\bar{k}_j(T) = \frac{\int_0^\infty dE_R \int_0^{J_{max}} dJ P_{C_4H_8^+}(E_R, J) k_j(E_R, J)}{\int_0^\infty dE_R \int_0^{J_{max}} dJ P_{C_4H_8^+}(E_R, J)}. \quad (III.23)$$

The lifetime of the complex is just the inverse of the sum of the three unimolecular rate constants

$$\tau(T) = 1 / \sum_{j=a,b,c} \bar{k}_j(T). \quad (III.24)$$

It is interesting to note that this unimolecular perspective can also be used to obtain an expression for the branching ratio which is equivalent to the expression in Eq. (III.19). This point is discussed in Appendix B.

A comparison of experimental and theoretical values for the product branching ratio, the reaction efficiency, and the complex lifetime is shown in Table I.

TABLE I. Branching ratios for the reactions

	Ion-Molecule reaction		Photoionization	
	Expt.	Theory ^b	Expt. ^c	Theory ^{b,c}
$C_2H_4^+ + C_2H_4$ or $C_4H_8 + h\nu \rightarrow (C_4H_8^+)$				
$\begin{matrix} \nearrow k_a & C_3H_5^+ + CH_3 \cdot \\ \searrow k_b & C_4H_7^+ + H \cdot \end{matrix}$				
k_b/k_a	0.11 ^a	0.13	0.38	0.38
Fraction converted	0.8 ± 0.2^a	0.64 ^d	...	...
Complex lifetime (s)	$2.9 \times 10^{-7}^e$	$2.9 \times 10^{-7}^f$	...	...

^aReference 11.^bThis work.^cReferences 8 and 53. See also Fig. 7. These numbers were obtained for an energy equal to the average thermal energy above the $C_2H_4^+ + C_2H_4$ threshold.^dThe fraction converted is calculated from Eq. (III.21) in this text.^eReference 32.^fCalculated from Eq. (III.24) in the text.

F. Translational energy release distributions

The translational energy release distributions for fragmentation of internally excited $C_4H_8^+$ have been determined using mass spectrometric techniques. The ions are created by electron impact on a thermal distribution of neutral C_4H_8 molecules according to the reaction



It is assumed, as it was for the photoionization processes in Sec. IIIB, that the thermal vibrational energy of the neutral is negligible, and that there is no change in the angular momentum or moment of inertia of the C_4H_8 species during the ionization process. Therefore, the angular momentum distribution function for the ion is given by the thermal distribution in Eq. (III.13). The vibrational energy distribution function corresponds to the energy deposition function for electron impact $P_{d\omega}(E_v)$, which can be estimated from the photoelectron spectrum.³⁵

These distribution functions correspond to ions as they are initially formed. Since the translational energy release distributions are determined from peak shapes for metastables which decompose in either the first or second field free region of a mass spectrometer, the additional factor

$$P_\tau(E_v, J) = \int_{\tau_1}^{\tau_2} k_{tot}(E_v, J) \exp[-k_{tot}(E_v, J)t] dt \quad (\text{III.26a})$$

$$= \exp[-k_{tot}(E_v, J)\tau_1] - \exp[-k_{tot}(E_v, J)\tau_2] \quad (\text{III.26b})$$

is needed to account for the fact that the number of ions which decompose in this region depends upon the time of entering the region τ_1 and the time of exiting τ_2 . These two times are determined from the experimental conditions.

The total probability that a $C_4H_8^+$ ion decomposes in the first or second field-free region with energy E and angular momentum J is then given by the product

$$P_{d\omega}(E_v)P(J)P_\tau(E_v, J) \quad (\text{III.27})$$

Multiplication by the factor

$$R_j(E_v, J) = k_j(E_v, J)/k_{tot}(E_v, J) \quad (\text{III.28})$$

gives the probability that fragmentation proceeds into product channel j . When this result is multiplied by $P_j(E_v, J; E_t)$, given by Eq. (II.23), the probability that fragmentation proceeds into product channel j with translational energy release E_t is obtained.

The total probability that the translational energy release in channel j is E_t is therefore given by

$$P_j(E_t) = \int_0^\infty dE P_{d\omega}(E_v) \int_0^\infty dJ P(J) P_\tau(E_v, J) R_j(E_v, J) P_j(E_v, J; E_t)$$

$$\int_0^\infty dE P_{d\omega}(E_v) \int_0^\infty dJ P(J) P_\tau(E_v, J) R_j(E_v, J) \quad (\text{III.29})$$

In this expression the numerator corresponds to the number of fragmentations in channel j with translational energy release E_t , while the denominator corresponds to the total number of fragmentations in channel j regardless of the translational energy release.

Equation (III.29) was used to calculate the translational energy release distributions for the $C_3H_5^+/CH_3 \cdot$ and $C_4H_7^+/H \cdot$ channels for an assumed ion source temperature of 473°K.³⁶ The resulting distributions are compared with the experimental curves in Figs. 8 and 9.

IV. DISCUSSION

A central question in the study of unimolecular reactions is the question of energy randomization or quasi-equilibrium in the unimolecular parent about to undergo reaction. The calculations reported here assume complete energy randomization in $C_4H_8^+$. This assumption is supported by the fact that all neutral isomers show nearly identical behavior upon ionization and reaction and by the current interpretation that the lifetime distribution of the $C_4H_8^+$ parent is essentially a pure exponential.⁷ However, the experimental behavior alone does not rule out the possibility that isolated states exist in $C_4H_8^+$ in the energy range probed by the experiments.³⁷ Hence, we would like to address this question in greater detail. In doing so it is instructive to discuss the $C_4H_8^+$ system concurrently with the $C_4H_6^+$ system^{2(a),38} since the behavior of the PIPECO absolute rate data and RRKM-QET calculations are very similar for both systems.

Experimental evidence for the existence of isolated vibrational states in metastable ions was first claimed by Werner and Baer^{38(a)} in a PIPECO study of the decay rate of $C_4H_6^+$. When these authors compared their experimental data to RRKM-QET calculations with the activation energy set equal to the zero-point energy difference between the most stable $C_4H_6^+$ structure and $C_3H_3^+ + CH_3 \cdot$, the theoretical results they obtained were several orders of magnitude lower than experiment

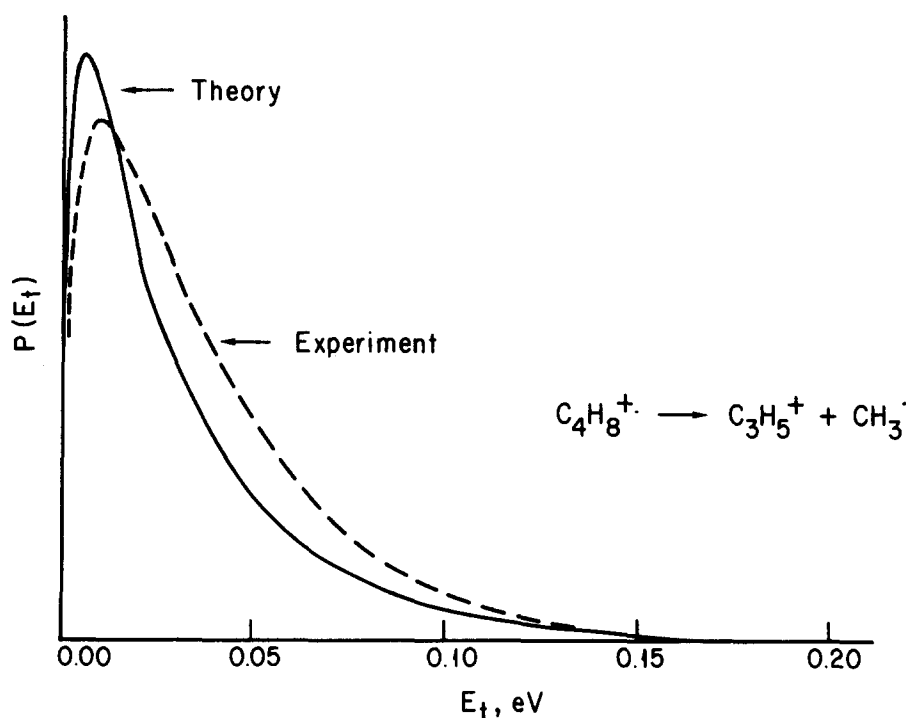
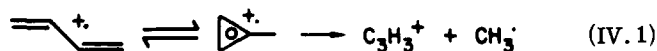


FIG. 8. Translational energy distribution in the $C_3H_5^+ + CH_3$ products resulting from unimolecular decomposition of $C_4H_8^+$. —, theory presented here; ----, experimental results (see Ref. 10).

(i.e., very similar to the results stated in Fig. 6 for $E_f^* = E_f^0$). They interpreted this behavior as evidence that isolated states exist in metastable $C_4H_8^+$ ions. The reasoning behind this interpretation is that isolated states will tend to decrease the density of states in the denominator of the rate constant expression [see Eq. (II.2)] and hence bring theory into agreement with experiment. However, Chesnavich and Bowers^{2(a)} noted that agreement between experiment and theory could also be obtained for $C_4H_8^+$ by increasing the reactive flux rather than decreasing the density. They accomplished this by taking the activation energy to be less

than the zero-point energy difference (the agreement they obtained between experiment and theory was similar to the switching model result plotted in Fig. 6). Chesnavich and Bowers interpreted this to imply that the main decomposition pathway of $C_4H_8^+$ is given by the following isomerization reaction:



In order to test this mechanism Baer and co-workers^{38(b)} subsequently measured the energy dependence of the

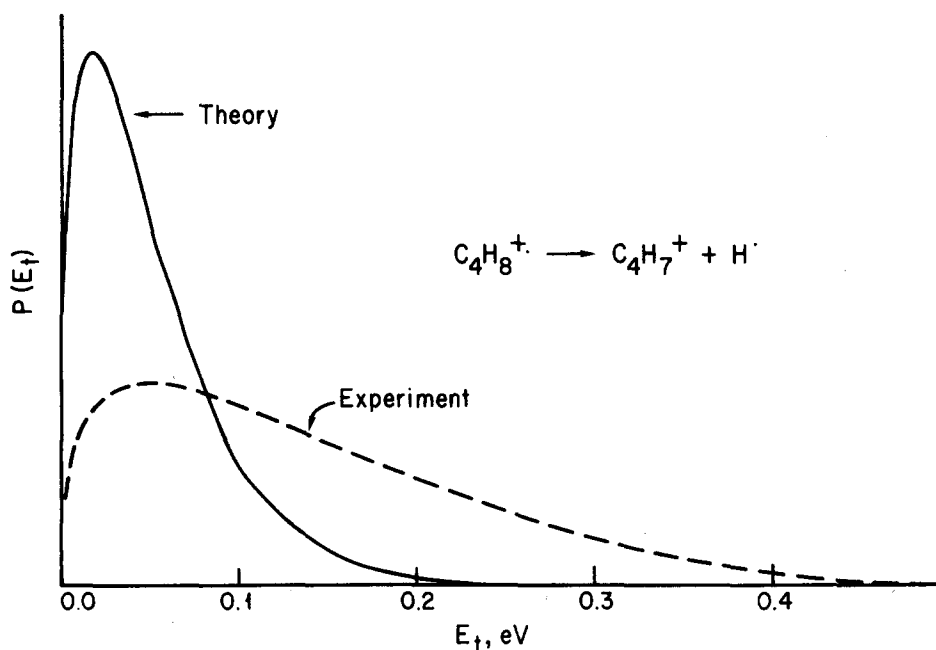


FIG. 9. Translational energy distribution in the $C_4H_7^+ + H$ products resulting from unimolecular decomposition of $C_4H_8^+$. —, Theory presented here; ----, experimental results (see Ref. 10).

methylcyclopropenium ion dissociation rate. They found it to be identical to that of the other $C_4H_5^+$ isomers. Hence, their data suggest that the isomerization interpretation offered by Chesnavich and Bowers to explain the choice $E^\ddagger < E^0$ is not correct.³⁹ On the other hand, the data do not disagree with the minimum flux interpretation presented here because the saddle point transition state controlling the dissociation rate may well lie to the product side of the methylcyclopropenium structure and still have an energy less than E_0 .

It is possible to now shed further light on this problem by estimating the effect isolated states in metastable $C_4H_5^+$ ions would have on the calculations reported in this paper. This can be accomplished by first lowering the density of states in the denominator of the unimolecular rate constant expressions in order to account for the isolated states. Then, to regain the agreement between experiment and theory for the unimolecular rate constants and branching ratios, the fluxes would also have to be lowered. However, making this latter change would increase the discrepancy between the experimental and theoretical bimolecular reaction efficiencies given in Table I. This problem arises because the unimolecular expressions depend on both the density of states of $C_4H_5^+$ and on the flux into each product channel whereas the bimolecular expressions depend only on the fluxes. (The bimolecular rate constant can also be written as a flux divided by a density. However, in this case the density refers to the $C_2H_4^+ + C_2H_4$ reactants rather than the $C_4H_5^+$ intermediate.) The problem can be reduced to some extent by lowering the $B^\ddagger$'s because the bimolecular results are much more sensitive to the $B^\ddagger$'s than the unimolecular results. However, the $B^\ddagger$ we obtained for the $C_3H_5^+$ channel is already near the lower limit expected from our estimated structure calculation (see Appendix D). Hence, it is the opinion of the present authors that large numbers of isolated states probably don't exist in metastable $C_4H_5^+$ and $C_4H_6^+$ ions, but rather that the dissociation dynamics is controlled primarily by minimum flux transition states for which the activation energies are less than the zero-point energy differences.

Further insight on this conflict can be obtained from both theoretical and experimental investigations of unimolecular dissociations. For example, Bunker, Hase, and their co-workers^{40,41} found activation energies less than the zero-point energy differences in their comparisons of the minimum density RRKM method to trajectory calculations of triatomic dissociations and to experimental data on the dissociation and recombination of ethane. Relevant experimental data can also be obtained from combined unimolecular-bimolecular studies of other small hydrocarbon ions (to avoid the problems associated with large kinetic shifts) in which the final step in the main dissociation pathway is a simple bond cleavage. Possible candidates include $C_3H_3^+$, $C_4H_{10}^+$, and the C_5 hydrocarbon ions. Data on the unimolecular rates and branching ratios and on the bimolecular efficiencies and branching ratios are all necessary in order to separate the effect isolated states would have on the density from the effect a minimum flux transition state with $E^\ddagger < E^0$ would have on the flux. Furthermore, it would be

useful to obtain the bimolecular data as a function of temperature in order to establish the $B^\ddagger$'s more precisely in the corresponding RRKM-QET calculations. Information on the translational and vibrational energy dependence of the bimolecular efficiencies and branching ratios⁴² could also be used. However, in these instances an additional assumption⁴³ is required to account for the dependence of the reaction on the initial energy distribution. Furthermore, high energy molecular beam experiments may lead to difficulties in interpretation⁴⁴⁻⁴⁶ because in these experiments the very high J distribution of the intermediate would drastically alter the system effective potential surface.

Within the context of variational TST the calculations reported here include, for practical reasons, the additional approximation that the location and properties of the tight transition states are taken to be independent of the total system energy and angular momentum. A more rigorous calculation would involve the determination of the point of minimum flux as a function of E and J .⁴⁷ In this sense the transition states used here can be viewed as average structures for the energy and angular momentum ranges sampled in the experiments.

It is possible to estimate the effect of this additional approximation by appeal to the fact that when TST fails, it does so in a way that provides an upper bound to the true reactive flux. For example, as the internal energy of the $C_4H_5^+$ parent ion increases, the relative abundance of $C_2H_4^+$ measured in the experiments of Meisels and co-workers⁸ slowly increases⁴⁸ to ~14% at 1.6 eV above the zero-point energy of $C_3H_5^+ + CH_3^+$. The calculated relative abundance, however, peaks at ~9% at 0.65 eV and then very gradually decreases to ~8% at 1.6 eV. This small difference in behavior can be explained by assuming that the parameters of the saddle point transition state between the deep well $C_4H_5^+$ complex and $(C_2H_4)_2^+$ dimer are essentially independent of energy and angular momentum, while, on the other hand, the parameters of the tight "minimum flux" transition states in the other channels depend on energy and angular momentum. If this is the case, then the parameters of the tight transition states used in this paper for the $C_3H_5^+$ and $C_4H_7^+$ channels will be most appropriate for the energy region studied by the PIPECO absolute rate experiments of Baer and co-workers, since this was the primary experimental data used to establish these parameters. [As stated in Appendix D, the ethylene ion-molecule reaction data were used to establish the rotational constants of the tight and saddle point transition states. However, the low (thermal) J distribution of neutral C_4H_6 makes the PIPECO data essentially independent of the rotational constants.] Hence, at higher energies the calculated fluxes through the tight transition states will represent upper bounds to the true values, whereas the flux through the saddle point transition state in the ethylene channel will remain approximately equal to the true value. This will, in turn, cause the relative abundances $C_3H_5^+$ and $C_4H_7^+$ to be slightly higher than the true values and that of $C_2H_4^+$ to be slightly lower, in agreement with the deviation between experiment and theory.

Thus, it appears that some of the discrepancies between experiment and theory can be attributed at least

in part to the additional approximation inherent in our calculations. Replacement of this approximation with a more rigorous treatment should tend to bring experiment and theory into better agreement. Unfortunately, this is not an easy task to accomplish.⁴⁹

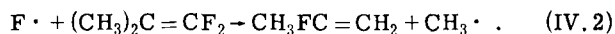
Recently, Rosenstock and co-workers⁵⁰ have combined variable residence time PIPECO experiments with RRKM-QET calculations in an effort to determine heats of formation of the daughter ions of metastable dissociations. Since heats of formation generally enter into the model presented here only through the orbiting transition states, it is useful to know to what extent the orbiting transition states influence the calculations. A cursory examination of our computer printout showed that $F^{\text{orb}} \gg F^\ddagger$ for energies greater than ~ 0.01 eV above the zero points of the separated fragments in the C_3H_5^+ and C_4H_7^+ channels and for energies greater than ~ 0.1 eV above the zero point of the separated fragments in the C_2H_4^+ channel. Hence, if the method of Rosenstock and co-workers was applied to the C_4H_8^+ system using the model presented here, the heat of formation of C_2H_4^+ should be most accessible to determination. In any case, the activation energies determined in this way lead to lower bounds on the heats of formation. Hence, combining the method of Rosenstock and co-workers with the model presented here should prove to be a useful complement to standard experimental methods of determining appearance potentials since the heats of formation obtained from these latter methods are upper bounds to the true values.

It should also be noted that RRKM-QET calculations of the dissociation rates of large hydrocarbons, such as the substituted benzene ions,^{1,2,50(a),51,52} tend to agree with PIPECO data when the activation energies are set equal to the zero-point energy differences. Whether this has to do with the larger kinetic shifts or, within the context of the transition state switching model, to the location of the minimum flux transition state is a question which can perhaps best be answered by studies such as the one reported here and those of Rosenstock and co-workers. Understanding this phenomenon better has important consequences for thermochemistry, because when the lower bound on a heat of formation obtained from the method of Rosenstock and co-workers agrees with the upper bound obtained from standard appearance potential measurements, then the most obvious conclusion is that the heat of formation has been established to a high degree of accuracy.

The assumption that the nonfixed energy is statistically partitioned at the orbiting transition state provides the basis for our calculation of the product translational energy distributions. When the orbiting transition state is rate determining, this assumption should certainly be valid. However, when the tight transition state is rate determining, this assumption implies that the net effect of the dynamics occurring between the two transition states is to maintain continual randomization of energy between all modes. This insures that the distribution of trajectories, which must be statistical at the rate determining tight transition state, remains statistical as the distribution propagates toward products. Failure

of this assumption to hold exactly may be the cause of some discrepancy between theory and experiment in the comparisons shown in Figs. 8 and 9.

As seen from these comparisons, the theoretical distributions appear to peak at lower energies and fall to zero more rapidly than the experimental data. For the C_3H_5^+ channel this disagreement is only slight, but for the C_4H_7^+ channel the experimental curve peaks at substantially higher energy than theory and falls to zero much slower. This result is very similar to the result obtained when the translational energy distribution calculated from the orbiting transition state is compared to the measured data for the reaction



Worley and Marcus²¹ obtained much better agreement between theory and experiment by explicitly taking into account the "transitional" modes that go from bending vibrations at the tight transition state to rotations in the products. Although the properties of this approach have not yet been studied for low J unimolecular reactions, we suggest that when applied to the translational energy distribution of C_4H_7^+ formed from low J C_4H_8^+ precursors the discrepancy between theory and experiment in Fig. 9 will diminish and perhaps disappear.

Another interesting set of experimental results are the C_3H_5^+ and C_4H_7^+ translational energy distributions which have recently been measured by Kemper and Bowers¹⁰ for C_4H_8^+ formed in the ZAB ion source from the ethylene ion-molecule reaction. In this case the C_4H_8^+ intermediate has very large total angular momentum relative to C_4H_8^+ ions formed by electron impact. Although a quantitative analysis of the results obtained will not be attempted in this paper, because the pressure and residence time in the ion source are not well characterized for these experiments, we offer the following qualitative explanation.

First, the measured C_3H_5^+ distribution shifts to higher translational energy than the unimolecular experimental result plotted in Fig. 8. We feel that this occurs because the large angular momentum of the C_4H_8^+ intermediates biases the statistical partitioning of the nonfixed energy of the products to higher values of the translational energy.

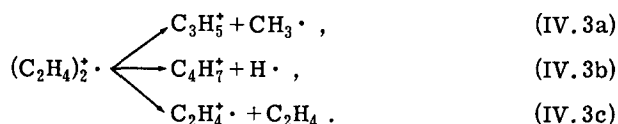
Second, there is very little difference in the C_4H_8^+ translational energy distribution measured for the ethylene ion-molecule reaction and the unimolecular result plotted in Fig. 9. Qualitatively, we feel the reason for this is that high angular momentum C_4H_8^+ ions preferentially react via the C_3H_5^+ channel rather than the C_4H_7^+ channel. This effect has previously been observed by Meisels and co-workers⁵³ and by Klots⁵⁴ and has been attributed to the decrease in flux in the C_4H_7^+ channel relative to that in the C_3H_5^+ channel because of the low mass and polarizability of the $\text{H} \cdot$ atom fragment. However, according to the model presented in this paper the orbiting transition states are not influential in the C_3H_5^+ and C_4H_7^+ channels at energies greater than the $\text{C}_2\text{H}_4^+ + \text{C}_2\text{H}_4$ zero-point energy. Rather, the calculations reported here suggest that high J C_4H_8^+ intermediates react preferentially to the C_3H_5^+ channel because the rota-

tional constant associated with the tight transition state in this channel ($B^\ddagger = 0.120 \text{ cm}^{-1}$) is smaller than the one associated with the tight transition state in the $\text{C}_4\text{H}_7^\ddagger$ channel ($B^\ddagger = 0.210 \text{ cm}^{-1}$). This difference in rotational constants allows higher J systems to pass through the tight transition state in the $\text{C}_3\text{H}_5^\ddagger$ channel than can pass through the tight transition state in the $\text{C}_4\text{H}_7^\ddagger$ channel.

Some interesting mechanistic insight can be gained by considering studies on isotopically labeled species. It has long been known that in low energy deuterated and ^{13}C labeled C_4H_8^+ ions complete scrambling occurs prior to dissociation.^{12(a)-(d)} However, several of the reactions of ethylene ions in $\text{C}_2\text{H}_4/\text{C}_2\text{D}_4$ mixtures, with ion energies ranging from thermal to several eV,^{12(e)} show only very small occurrences (1%–2%) of the isotopically scrambled $\text{C}_2\text{H}_3\text{D}^+$, $\text{C}_2\text{H}_2\text{D}_2^+$, and C_2HD_3^+ ions. Reaction of low energy C_2H_4^+ ions with C_2D_4 results in large C_2D_4^+ signals, accounting for 20%–30% of the total product ion intensity.

Calculations based on the model presented here indicate that for the thermal energy ethylene ion–molecule reaction 36% of all collision pairs which reach the shallow $(\text{C}_2\text{H}_4)_2^+$ van der Waals type well will back dissociate to $\text{C}_2\text{H}_4^+ + \text{C}_2\text{H}_4$ rather than proceed to the deep C_4H_8^+ well. On the other hand, of those systems which do reach the deep C_4H_8^+ well, only 6% are predicted to undergo back reaction to the $\text{C}_2\text{H}_4^+ + \text{C}_2\text{H}_4$ channel. Assuming that this behavior is also followed, at least qualitatively, by the $\text{C}_2\text{H}_4^+ + \text{C}_2\text{D}_4$ system, we suggest that the relatively large C_2D_4^+ intensity observed in the low energy $\text{C}_2\text{H}_4^+ + \text{C}_2\text{D}_4$ reaction results from collision pairs which back dissociate from the shallow $\text{C}_2\text{H}_4(\text{C}_2\text{D}_4)^+$ well. It is highly unlikely that any H/D scrambling occurs on this part of the potential energy surface since the two ethylene species are only loosely bound. The very small $\text{C}_2\text{H}_3\text{D}^+$, $\text{C}_2\text{H}_2\text{D}_2^+$, and C_2HD_3^+ intensities probably result from the small fraction of collision pairs which actually reach the deep $\text{C}_4\text{H}_8\text{D}_4^+$ well, where rapid H/D scrambling may occur, and then back dissociate into the ethylene channel.

Another point which should be addressed here is the recent study reported by Ceyer *et al.*²³ on the photoionization of the neutral van der Waals dimer $(\text{C}_2\text{H}_4)_2$. These authors reported appearance potentials for the reactions



The appearance potentials they measured for Reactions (IV. 3b) and (IV. 3c) are in agreement with accepted thermodynamic thresholds for these reactions. The appearance potential for Reaction (IV. 3a), however, is 7.7 kcal/mol greater than the accepted thermodynamic threshold.^{6,55} Ceyer *et al.* interpreted this result to imply that there is a barrier of this magnitude in the exit channel for formation of C_3H_5^+ .

This interpretation is in conflict with the data and analysis presented in this paper and with the appearance potentials obtained from photoionization of the various

C_4H_8 neutral isomers. For example, the measured translational energy distribution for C_3H_5^+ plotted in Fig. 8 rises rapidly from zero, peaks at $\sim 0.01 \text{ eV}$, and then falls smoothly to zero. This distribution is characteristic of a system that does not have a substantial barrier to the reverse reaction. Furthermore, if such a barrier did exist, it would be extremely difficult, if not impossible, to simultaneously fit the PIPECO absolute unimolecular rate and branching ratio data using the model presented in this paper or, for that matter, any other reasonable model.

The results of Ceyer *et al.* for the C_3H_5^+ channel are, to us, puzzling and lend themselves to no apparent interpretation at this time. What is apparent is that the data and analysis presented in this paper strongly suggest that the postulated barrier of 7.7 kcal/mol is not a correct interpretation.

Finally, an important point to keep in mind is that all adjustable parameters used in the calculations were determined on the basis of the unimolecular decay rates, the unimolecular branching ratio, and the bimolecular branching ratio. The bimolecular reaction efficiency and the lifetime of the bimolecular complex given in Table I, as well as the product translational energy distributions in Figs. 8 and 9, were all calculated with essentially no adjustable parameters. The fact that all of these results are either in good agreement with experiment or, in the case of the $\text{C}_4\text{H}_7^+ + \text{H} \cdot$ translational energy distribution, can be explained within the context of the theory strongly supports the validity of the model presented in this paper.

APPENDIX A

In this Appendix we first summarize the derivation of Miller's UST reaction probability [Eq. (II. 5)] and then show how Miller's fundamental assumption must be altered to obtain the double deep well probability [Eq. (II. 22)].

To obtain the UST expression we consider a potential surface for which the following extrema in the local flux exist, in order, along the reaction coordinate between reactants and products: (1) a minimum flux F_1 through a surface S_1 , (2) a maximum flux F_x through a surface S_x , and (3) a second minimum flux F_2 through a surface S_2 . The fundamental assumption of UST is as follows⁴: The probability that a trajectory, reaching S_x from the reactant side, goes directly out S_2 to products is

$$P(x \rightarrow 2) = F_2/F_x. \quad (\text{A1})$$

Likewise, the probability that a trajectory, reaching S_x from the products side, goes directly out S_1 to reactants is

$$P(x \rightarrow 1) = F_1/F_x. \quad (\text{A2})$$

To obtain the UST reaction probability consider a trajectory that has just crossed S_1 from reactants. The probability that it goes directly out S_2 to products is $P(x \rightarrow 2)$. The probability that it crosses S_x but turns, recrosses S_x , and goes directly out S_1 to reactants is $[1 - P(x \rightarrow 2)]P(x \rightarrow 1)$. The probability that, rather than going directly out S_1 to reactants, it instead turns, re-

crosses S_x another time, and goes directly out S_2 to products is $[1 - P(x-2)][1 - P(x-1)]P(x-2)$. Hence, if n is the number of additional times after the first that a trajectory crosses S_x from the reactant side, then the total reaction probability is given by

$$\omega = \sum_{n=0}^{\infty} [1 - P(x-2)]^n [1 - P(x-1)]^n P(x-2). \quad (\text{A3})$$

Performing the summation⁵⁶ gives Eq. (II.5).

The same arguments can be used to determine ω' , the probability that the trajectory doesn't react. In this case

$$\omega' = \sum_{n=0}^{\infty} [1 - P(x-2)]^{n+1} [1 - P(x-1)]^n P(x-1), \quad (\text{A4a})$$

$$= \frac{F_1 - F_1 F_2 / F_x}{F_1 + F_2 - F_1 F_2 / F_x}. \quad (\text{A4b})$$

Note that $\omega + \omega' = 1$. Hence, the total flux through S_1 from reactants is conserved.

To obtain the double, deep well reaction probability [Eq. (II.22)] we consider a potential surface for which two deep wells W_a and W_b exist in order along the reaction coordinate between reactants and products. In this case S_1 is a surface of minimum flux F_1 separating the reactants from W_a , S_x is a surface of minimum flux separating W_a and W_b , and S_2 is a surface of minimum flux separating W_b from the products. The probabilities of passage from W_a to W_b and from W_b to W_a are taken to be

$$P(a-b) = F_x / (F_x + F_1) \quad (\text{A5})$$

and

$$P(b-a) = F_x / (F_x + F_2). \quad (\text{A6})$$

Likewise, the probabilities of passage from W_b to products and from W_a to reactants are taken to be

$$P(b-2) = F_2 / (F_2 + F_x) \quad (\text{A7})$$

and

$$P(a-1) = F_1 / (F_1 + F_x). \quad (\text{A8})$$

To obtain the reaction probability consider a trajectory that has just crossed S_1 from reactants to W_a . The probability that it crosses S_x to W_b and then crosses S_2 to products is $P(a-b)P(b-2)$. The probability that, rather than going out S_2 to products, it recrosses S_x into W_a , recrosses again back to W_b , and then crosses S_2 to products is $P^2(a-b)P(b-a)P(b-2)$. Hence, if n is the number of additional times after the first that a trajec-

tory crosses S_x from W_a to W_b , then the total reaction probability is given by

$$\omega = \sum_{n=0}^{\infty} P^{n+1}(a-b)P^n(b-a)P(b-2). \quad (\text{A9})$$

Performing the summation⁵⁶ gives Eq. (II.22). The same arguments, using Eq. (A8), can be used to determine ω' , the probability that the trajectory doesn't react. This leads to $\omega' = 1 - \omega$. Hence, the total flux through S_1 from reactants is conserved.

The essential difference between UST and the double, deep well model is that the probabilities given by Eqs. (A1) and (A2) assume that the dynamics in the interaction region is direct, whereas the probabilities given by Eqs. (A5)–(A8) assume that the dynamics is statistical.

APPENDIX B

For the bimolecular ion–molecule reaction in Eq. (III.17), the unimolecular rate constant for decomposition of chemically activated $C_4H_8^+$ into channel j is given by Eq. (III.23). The ratio of these unimolecular rate constants for channels a and b must then yield the appropriate ratio of products:

$$\frac{[C_4H_7^+]}{[C_3H_5^+]} = \frac{\bar{k}_b(T)}{\bar{k}_a(T)}. \quad (\text{B1})$$

In this Appendix we prove that Eq. (B1), derived in terms of the unimolecular reaction of chemically activated $C_4H_8^+$, is equivalent to Eq. (III.19), derived in terms of the bimolecular ethylene ion–molecule reaction.

When Eq. (III.23) is substituted into Eq. (B1) the result is

$$\frac{[C_4H_7^+]}{[C_3H_5^+]} = \frac{\int_0^\infty dE_R \int_0^{J_{\max}} dJ P_{C_4H_8^+}(E_R, J) k_b(E_R, J)}{\int_0^\infty dE_R \int_0^{J_{\max}} dJ P_{C_4H_8^+}(E_R, J) k_a(E_R, J)}. \quad (\text{B2})$$

Using Eq. (III.22) for $P_{C_4H_8^+}(E_R, J)$, then expanding $k_{\text{tot}}(E_R, J)$ in terms of the separate $k_j(E_R, J)$, and using Eqs. (III.4) for the $k_j(E_R, J)$, Eq. (B2) becomes

$$\frac{[C_4H_7^+]}{[C_3H_5^+]} = \frac{\int_0^\infty dE_R \exp(-E_R/k_B T) \int_0^{J_{\max}} dJ 2J F'_a(F'_b / (F'_a + F'_b + F'_c))}{\int_0^\infty dE_R \exp(-E_R/k_B T) \int_0^{J_{\max}} dJ 2J F'_a(F'_a / (F'_a + F'_b + F'_c))}. \quad (\text{B3})$$

Substitution for F'_c from Eq. (III.5c) and subsequent rearrangement yields

$$\frac{[C_4H_7^+]}{[C_3H_5^+]} = \frac{\int_0^\infty dE_R \exp(-E_R/k_B T) \int_0^{J_{\max}} dJ 2J F_c^{\text{orb}}(F'_b / (F'_a + F'_b + F_c^{\text{orb}}) + [(F'_a + F'_b) F_c^{\text{orb}}] / F_c^{\text{orb}})}{\int_0^\infty dE_R \exp(-E_R/k_B T) \int_0^{J_{\max}} dJ 2J F_c^{\text{orb}}(F'_a / (F'_a + F'_b + F_c^{\text{orb}}) + [(F'_a + F'_b) F_c^{\text{orb}}] / F_c^{\text{orb}})}, \quad (\text{B4})$$

which is identical to Eq. (III.19). Thus, the bimolecular and unimolecular perspectives on the chemical activation system lead to identical results, as they should.

It is interesting to compare Eqs. (B3) and (B4). Each contains the product of a flux and a fractional flux. In

Eq. (B3), which embodies the unimolecular perspective, the flux F'_c represents the rate of passage of reactant pairs to the deep $C_4H_8^+$ well. The corresponding fraction represents the probability that a system which originates at the deep $C_4H_8^+$ well will decompose into a particular product channel. In Eq. (B4), which embodies

the bimolecular perspective, the flux F_c^{orb} represents the rate of passage of reactant pairs to the loosely bound $(\text{C}_2\text{H}_4)_2^+$ dimer well. The corresponding fraction represents the probability that a system which originates at the $(\text{C}_2\text{H}_4)_2^+$ well will decompose into a particular product channel. Either perspective yields the correct result.

APPENDIX C

This Appendix summarizes the expressions used to evaluate the various densities and fluxes. All quantities are expressed in terms of E , the vibrational-rotational energy of the deep well C_4H_8^+ complex (see Fig. 5). They can be rewritten in terms of E_v and E_R by using Eqs. (III. 7c) and (III. 9b).

1. The transition state fluxes

The total flux at a tight transition state can be written as

$$F^\ddagger(E, J) = N_v^\ddagger(x) 2J/h, \quad (\text{C1})$$

where $N_v^\ddagger(x)$ is the sum of states for the vibrational modes of the transition state at vibrational energy less than or equal to x , where

$$x = E - E^\ddagger - B^\ddagger J^2. \quad (\text{C2})$$

In Eq. (C1) the transition state structure is assumed to be that of a spherical top. Hence, the $2J$ degeneracy accounts for the projection of J on an internal axis of the top. Since $E^\ddagger \leq E^0$, Eq. (C1) should also include the step function $h(E - E^0)$ to insure that $F^\ddagger > 0$ only when there is enough energy available for reaction to occur. However, the switching model accounts for this in a natural way through Eqs. (III. 5a)–(III. 5c). Therefore, the step function has been dropped in Eq. (C1).

The total flux at the orbiting transition state can be written as

$$F^{\text{orb}}(E, J) = \int_{E_{\text{tr}}^\ddagger}^{E-E^0} \rho_v(x) \Gamma_{\text{ro}}(E_{\text{tr}}, J) dE_{\text{tr}}/h, \quad (\text{C3})$$

where $\Gamma_{\text{ro}}(E_{\text{tr}}, J)$ is the sum of rotational-orbital states at rotational-translational energy E_{tr} ,⁵⁷ and where $E_{\text{tr}}^\ddagger$ is the minimum E_{tr} for which $\Gamma_{\text{ro}}(E_{\text{tr}}, J) > 0$.⁵⁸ The density $\rho_v(x)$ is the density of states of the vibrational modes of the transition state pair at vibrational energy x , where x now equals

$$x = E - E_0 - E_{\text{tr}}. \quad (\text{C4})$$

The equations for $\Gamma_{\text{ro}}(E_{\text{tr}}, J)$ have been given previously^{46,58} and have been clearly summarized by Verboom and Meisels.⁵⁹

If the transition state pair consists of two three-dimensional rotors, or a three-dimensional rotor and an atom, then it is possible, within the spherical top approximation, to simplify Eq. (C3). This leads to⁶⁰

$$F^{\text{orb}}(E, J) = \int_0^{L^+} N_{\text{vr}}(X_+) dL/h - \int_0^{L_-} N_{\text{vr}}(X_-) dL/h, \quad 0 \leq J \leq J_1, \quad (\text{C5})$$

$$= \int_{L_-}^{L^+} N_{\text{vr}}(X_-) dL/h, \quad J_1 \leq J \leq J_{\text{max}}, \quad (\text{C6})$$

where

$$X_\pm = E - E^0 - B_r(L \pm J)^2 - L^4/\lambda. \quad (\text{C7})$$

If the angular momenta are written in units of $\hbar = h/2\pi$, then $\lambda = 8\mu^2 q^2 \alpha / \hbar^4$ for an ion-neutral pair, where μ is the reduced mass of the pair and q the charge on an electron. The limits $L_\pm$ are given by the values of L that solve $X_\pm = 0$ if $0 \leq J \leq J_1$, and by the two values that solve $X_- = 0$ if $J_1 \leq J \leq J_{\text{max}}$.

The quantities B_r and N_{vr} depend on the rotational structure of the transition state pair. If the pair consists of two spherical top molecules, then $B_r^\ddagger = B_1^\ddagger + B_2^\ddagger$, where B_1 and B_2 are the individual rotational constants. In this case N_{vr} is the sum of states of a collection of vibrational and rotational degrees of freedom corresponding to the vibrational modes of the pair, a pair of two-dimensional rotors with rotational constants B_1 and B_2 , and a one-dimensional rotor with rotational constant $B_1 + B_2$. If the pair consists of a spherical top molecule and an atom, then B_r is the rotational constant of the molecule. In this case N_{vr} is the sum of states of a collection of vibrational and rotational degrees of freedom corresponding to the vibrational modes of the molecule and a two-dimensional rotor with rotational constant B_r . A conceptual explanation of the form of N_{vr} is given below.

The angular momentum boundaries J_1 and J_{max} are given by

$$J_1 = \sqrt{(E - E^0)/B_r}, \quad (\text{C8})$$

and³⁴

$$J_{\text{max}} = L^\ddagger + 2L^\ddagger^3/B_r\lambda, \quad (\text{C9})$$

where $L^\ddagger$ is given by the solution to

$$4L^\ddagger^6/B_r\lambda^2 + L^\ddagger^4/\lambda + E^0 - E = 0. \quad (\text{C10})$$

When C_4H_8^+ is formed from collisions between C_2H_4^+ and C_2H_4 an additional restriction³⁴ on J_{max} arises from the fact that at a given E the rotational energy of the C_4H_8^+ complex must be less than or equal to E . Hence, for the bimolecular experiments the auxiliary condition

$$J_{\text{max}} = \sqrt{E/B} \quad (\text{C11})$$

also applies. The actual value of J_{max} is then given by the lesser of Eqs. (C9) and (C11). Also, conservation of flux implies that trajectories with a given value of J must be able to pass through both the tight and orbiting transition states in a given channel. However, the switching model accounts for this in a natural way through Eqs. (III. 5a)–(III. 5c).

Finally, we note that Eqs. (C1), (C3), (C5), and (C6) refer to the flux at fixed J and $J_\pm$, where $J_\pm$ is the projection of J on a space fixed axis. This fact leads to the following conceptual explanation for the form of N_{vr} . Consider, for example, the orbiting flux for a pair of spherical top molecules. In this case there are a total of nine rotational-translational degrees of freedom in the center of mass system: three translational and six rotational. However, in $F^{\text{orb}}(E, J)$ these degrees of freedom are subject to the following constraints: (1) J is conserved, (2) $J_\pm$ is conserved, and (3) the radial mo-

tion of the pair is constrained to the orbiting transition state. Hence, of the nine degrees of freedom only an "effective" six enter into $F^{\text{orb}}(E, J)$. These effective degrees of freedom correspond in N_{vr} to the pair of two-dimensional rotors, the one-dimensional rotor, and the integration of N_{vr} over L . Similar arguments apply for a spherical top-atom pair.

II. The unimolecular rate constant

By summing the reactive flux at fixed E , J , and J_z over all available J_z , one obtains the *total* reactive flux at fixed E and J , i. e., the total reactive flux is given by $2JF(E, J)$, where, depending on the properties of the reaction channel under consideration, F equals F^{orb} , $F^\ddagger$, or $F^\ddagger\omega$. If we assume that the unimolecular parent is a spherical top, and if $k(E, J)$ is the unimolecular rate constant at fixed E , J , J_z , and K , where K is the projection of J on an internal axis of the parent, then the total reactive flux also equals $(2J)^2 \rho_v(E - BJ^2) k(E, J)$, where the $2J$ factors arise from summing over J_z and K , and where $\rho_v(E - BJ^2)$ is the vibrational density of states of the parent at vibrational energy $E_v = E - BJ^2$. Hence,

$$k(E, J) = F(E, J) / 2J \rho_v(E - BJ^2). \quad (\text{C12})$$

Note that $k(E, J)$ is defined at fixed J_z and K . Therefore, when averaged over a thermal distribution in J the distribution function must have a factor of $(2J)^2$ to account for J_z and K . That is why we have included factors of $(2J)^2$ in the numerator and denominator of Eq. (III.13). If the parent happens to be a linear molecule, then arguments analogous to those leading to Eq. (C12) show that the $2J$ factor is dropped from the denominator of Eq. (C12), and that the $(2J)^2$ factor is replaced by $2J$ in the numerator and denominator of Eq. (III.13).

III. The orbiting flux per unit translational energy

For a pair of spherical tops, and for a spherical top molecule and an atom, the orbiting transition state flux per unit translational energy $F^{\text{orb}}(E, J; E_t)$ can be written as⁶⁰

$$F^{\text{orb}}(E, J; E_t) = 2 \int_0^{y_1} \rho_{vr}(x) dy \quad (\text{C13})$$

if

$$J_r^* + J \leq L^*, \quad (\text{C14})$$

and

$$F^{\text{orb}}(E, J; E_t) = 2 \int_0^{y_1} \rho_{vr}(x) dy - \int_{L^* - J}^{y_2} \rho_{vr}(x) dy \quad (\text{C15})$$

if

$$|J_r^* - J| \leq L^* \leq J_r^* + J. \quad (\text{C16})$$

In these equations, ρ_{vr} is the density of states corresponding to N_{vr} , and the energy x is now given by

$$x = E - E^0 - E_t - B_r y^2; \quad (\text{C17})$$

also,

$$y_1 = \text{lesser of } \{J^*, J\}, \quad (\text{C18})$$

$$y_2 = \text{lesser of } \{J^*, L^* + J\}, \quad (\text{C19})$$

and

$$J^* = \sqrt{(E - E^0 - E_t) B_r}, \quad (\text{C20})$$

$$L^* = (\lambda E_t)^{1/4}; \quad (\text{C21})$$

also, $F^{\text{orb}}(E, J; E_t) = 0$ when the conditions $J^* < J$ and $L^* < J - J^*$ simultaneously hold.

APPENDIX D

There are seven species involved in the calculations presented here: the C_4H_8^+ ion (corresponding to the deep well), and the tight and orbiting transition states in each of the three product channels. For each of these species values are needed for the fundamental frequencies of each vibrational degree of freedom, the rotational constant for each (internal and external) rotational degree of freedom, and the difference in zero-point energy with the C_4H_8^+ ion.

Each of the orbiting transition states was treated as the separated ion and molecule. For each of these and the C_4H_8^+ ion all parameters could be estimated fairly reliably from literature values.^{61,62} Vibrational frequencies, when not available directly, were estimated from values for similar modes in corresponding compounds. Rotational constants were calculated from structural information.^{62,63} For our purposes here the values 1.09, 1.54, and 1.34 Å were used as the C-H, C-C, and C=C bond lengths, respectively. All bond angles were assumed to correspond to either planar or tetrahedral structures. The C_3H_5^+ and C_4H_7^+ ions were assumed to have the structures I and II

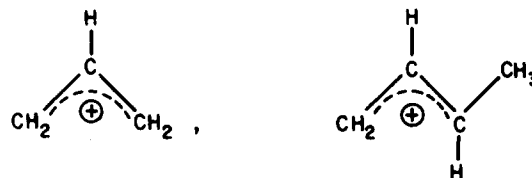


TABLE II. Parameters for C_4H_8^+ and tight transition states.

	C_4H_8^+	$\text{C}_3\text{H}_5^+ \cdots \text{CH}_3$	$\text{C}_4\text{H}_7^+ \cdots \text{H}$	$\text{C}_2\text{H}_4^+ \cdots \text{C}_4\text{H}_4$
ν_i^a	3100(3) 3000(5) 1700(2) 1400(5) 1200(2) 1100(3) 1000(1) 950(3) 900(2) 400(1) 300(2) 200(1)	3100(3) 3000(5) 1480(1) 1400(2) 1390(1) 1200(2) 1140(1) 1080(1) 950(2) 910(2) 570(2) 530(2) 470(1) 450(1) 400(1) 370(2)	3100(3) 3000(4) 1700(2) 1400(4) 1200(2) 1100(2) 1000(1) 950(3) 930(1) 900(2) 730(1) 400(1) 300(2) 200(1)	3100(3) 3000(5) 1700(2) 1400(5) 1200(2) 1100(3) 950(3) 900(2) 400(1) 300(2) 200(1)
σ^b	2	1	1	2
$B^{\ddagger c}$	0.255	0.120	0.210	0.120
$E_t^{\ddagger d}$		2.025	2.045	2.020

^aVibrational frequencies in cm^{-1} (degeneracies in parentheses).

^bSymmetry number.

^cRotational constant in cm^{-1} .

^dZero-point energy difference with C_4H_8^+ in eV.

TABLE III. Parameters for orbiting transition states.

	$C_3H_5^+ + CH_3 \cdot$	$C_4H_7^+ + H \cdot$	$C_2H_4^+ \cdot + C_2H_4$
ν_i^a	3100(4)	3100(3)	3100(4)
	3000(4)	3000(4)	3000(4)
	1440(1)	1700(2)	1620(2)
	1400(2)	1400(4)	1440(2)
	1380(1)	1200(2)	1340(2)
	1200(2)	1100(2)	1240(2)
	1150(2)	1000(1)	1020(2)
	950(2)	950(3)	950(4)
	910(2)	900(2)	830(2)
	650(2)	400(1)	
	480(1)	300(2)	
	400(1)	200(2)	
B_1^b	0.539	0.264	1.588
B_2^b	7.560	...	1.588
E^0^c	2.255	2.173	2.580
α^d	2.20	0.667	4.22

^aVibrational frequencies in cm^{-1} (degeneracies in parentheses).

^bRotational constants of fragments.

^cZero-point energy difference with $C_4H_8^+$ in eV (assumed = ΔH_f).

^dPolarizability of neutral in $\text{\AA}^3$.

respectively. The $C \equiv C$ bond lengths were taken to be halfway between a single and double bond, or 1.43 $\text{\AA}$. The E_j^0 values were taken to be the difference in heats of formation (at 298 K) between the products and the $C_4H_8^+$ ion:

$$E_j^0 = \Delta H_{f,298}(j) - \Delta H_{f,298}(C_4H_8^+) \quad (D1)$$

Ideally, $\Delta H_{f,0}$ values are preferable, but insufficient availability of these values led to the decision to use $\Delta H_{f,298}$ values consistently throughout. Equation (D1) also ignores the $P\Delta V$ term which would contribute only about 0.04 eV and is essentially negligible in the type of study reported here. Hence, for $C_4H_8^+$ and the three orbiting transition states all parameters were chosen on a rational basis without being adjusted according to the results of the calculations.

For the tight transition states the vibrational frequencies were estimated as being somewhere in between those for the $C_4H_8^+$ ion and the separated fragments. Somewhat arbitrarily, the $C_3H_5^+/CH_3 \cdot$ tight transition state frequencies were assumed to lie about $\frac{2}{3}$ of the way towards the product frequencies, while for the other two channels the tight transition state frequencies were assumed to lie about $\frac{1}{3}$ of the way towards the product frequencies. This leaves as adjustable parameters the zero-point energy difference E_j^0 and rotational constants B_j^0 for the tight transition states. The E_j^0 values were chosen to be less than or equal to the corresponding E_j^0 values. Reasonable limits were set on the B_j^0 values by treating each tight transition state as a normal molecule with one elongated bond (the reaction coordinate). The rotational constant was calculated for a normal bond length along the reaction coordinate and for a bond length of 5 $\text{\AA}$. This latter number is probably a reasonable upper limit at the tight transition state. Thus, although

the E_j^0 and B_j^0 values were adjustable, some limits did exist to insure that they were, in fact, reasonable.

The E_j^0 values have the most noticeable effect on the low- J unimolecular PIPECO calculations, while the B_j^0 values are predominant in the bimolecular cases. Keeping this in mind, an initial set of B_j^0 was assumed, and then the E_j^0 were adjusted to give the best fit to the experimental PIPECO (low J) results. These E_j^0 values were then kept constant and the B_j^0 values adjusted to give the best fit to the bimolecular branching ratio data. This procedure was repeated a number of times until all values remained essentially unchanged. The final values were then used to calculate all of the rate constants and the translational energy release distributions and the lifetime of the bimolecular intermediate.

A summary of the final parameters used is shown in Tables II and III.

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- ²⁸The classical approximation for rotational energy is assumed throughout.
- ²⁹(a) See, for example, the chapter by T. Baer in Ref. 1. (b) The inability to fit an arrival time distribution curve using a single exponential decay rate may indicate either that the J dependence of the rate constant is substantial or that more than one mechanism is involved in the fragmentation. In the present case a single exponential decay rate can fit the data (Ref. 7) so a problem doesn't arise. (c) It is of interest to note that a thermodynamic analog of the method used to derive Eq. (III.15) arises in the comparison of the average energy approach for calculating ion-dipole capture rates given by R. A. Barker and D. P. Ridge, *J. Chem. Phys.* **64**, 4411 (1976) to the microcanonical variational TST approach given in Refs. 25 and 26 [W. J. Chesnavich and M. T. Bowers (unpublished results)].
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